9. Develop a Student Management System using Spring MVC Framework, Dependency Injection (DI), JavaBeans, Controllers, JDBC Integration, and Annotation-based Configuration.

Index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Student Registration</title>
</head>
<body>
   <h2>Add New Student</h2>

  <form action="save" method="post">
      <label>Name:</label><br>
      <input type="text" name="name" required><br><br>

      <label>Course:</label><br>
      <input type="text" name="course" required><br><br>

      <button type="submit">Submit</button>
   </form>
</body>
</html>
```

Student.java

```
package com.example;

public class Student {
   private int id;
   private String name;
   private String course;
```

```
    public Student() {}
    public Student(int id, String name, String course) {
        this.id = id;
        this.name = name;
        this.course = course;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public String getCourse() { return course; }
    public void setCourse(String course) { this.course = course; }
}
```

StudentDao.java

```
package com.example;

import org.springframework.stereotype.Repository;
import java.sql.*;
import java.util.*;

@Repository
public class StudentDao {

    private Connection getConnection() throws SQLException {
```

```
        return DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb", "root", "password");
    }

    public void save(Student s) {
        String sql = "INSERT INTO student (name, course) VALUES (?, ?)";
        try (Connection con = getConnection(); PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setString(1, s.getName());
            ps.setString(2, s.getCourse());
            ps.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }

    public List<Student> getAll() {
        List<Student> list = new ArrayList<>();
        try (Connection con = getConnection();
             Statement st = con.createStatement();
             ResultSet rs = st.executeQuery("SELECT * FROM student")) {
            while (rs.next()) {
                list.add(new Student(rs.getInt("id"), rs.getString("name"), rs.getString("course")));
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return list;
    }
}
```

StudentController.java

```
package com.example;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.*;
import java.util.List;

@Controller
public class StudentController {

    @Autowired // Dependency Injection
    private StudentDao dao;

    @PostMapping("/save")
    public String saveStudent(@ModelAttribute Student student) {
        dao.save(student);
        return "redirect:/list";
    }

    @GetMapping("/list")
    @ResponseBody
    public List<Student> listStudents() {
        return dao.getAll();
    }
}
```

AppConfig.java

```
package com.example;
```

```
import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.Configuration;
import org.springframework.web.servlet.config.annotation.*;

@Configuration
@EnableWebMvc
@ComponentScan("com.example")
public class AppConfig implements WebMvcConfigurer {

    @Override
    public void configureDefaultServletHandling(DefaultServletHandlerConfigurer configurer)
{
        configurer.enable();
    }
}
```